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EXPERIEMNT 22

Prolog Program: Bird Can Fly or Not

Aim

To write a Prolog program to determine whether a particular bird can fly or not and incorporate the required queries.

Algorithm

1. Define facts for birds that can fly.
2. Define facts for birds that cannot fly.
3. Create a rule can_fly/1 to determine whether a bird can fly.
4. Give the bird name as a query.
5. Prolog checks the facts and displays whether the bird can fly or not.

Prolog Code

```
% Bird facts


bird(parrot).

bird(eagle).

bird(pigeon).

bird(crow).

bird(ostrich).

bird(penguin).
```

```
% Birds that can fly

flies(parrot).
flies(eagle).
flies(pigeon).
flies(crow).

% Birds that cannot fly

cannot_fly(ostrich).
cannot_fly(penguin).

% Rules

can_fly(Bird) :-
   bird(Bird),
   flies(Bird).

cannot_fly_bird(Bird) :-
   bird(Bird),
   cannot_fly(Bird).
```

Input / Required Queries

```
can_fly(eagle).
```

Output:

```
| ?- consult('C:/Users/MOSHIYA/Downloads/exp 22.pl').
compiling C:/Users/MOSHIYA/Downloads/exp 22.pl for byte code...
C:/Users/MOSHIYA/Downloads/exp 22.pl compiled, 29 lines read - 1620 bytes written, 16 ms

yes
| ?- can_fly(eagle).

yes
| ?- can_fly(penguin).

no
| ?- |
```

Result

Thus, the Prolog program was successfully implemented to determine whether a particular bird **can fly or cannot fly** using facts, rules, and appropriate queries.